VP10 — Datagram encoding and packets

Revision: `b3011e73ee1eece4b0df6861ea14435f312f5e54fcccec544f2b0ceebfe7012e`.

Only closed contracts with validated Encoding and explicit bit positions.

Packet: PacketBig / PacketBody — big-endian, most-significant-first

Source facts: f0049, f0050, f0051, f0052, f0053, f0054, f0055, f0056, f0057, f0058, f0065, f0066, f0067, f0068, f0080, f0081, f0082, f0083.

Declarations: PacketFlag, PacketNumber, PacketPayload.

Packet: PacketLittle / PacketBody — little-endian, least-significant-first

Source facts: f0055, f0056, f0057, f0058, f0059, f0060, f0061, f0062, f0063, f0064, f0065, f0066, f0067, f0068, f0080, f0081, f0082, f0083.

Declarations: PacketFlag, PacketNumber, PacketPayload.

Scope and notation

Each packet is one validated fixed Encoding of a Variant in a closed Contract. Bit offsets are logical contract positions; byte-order and bit-order are explicit source declarations. The view does not implement serialization or infer padding, variable lengths, optional-field layout or an executable codec. Header fields belong to the Contract; variant fields belong to the Variant. Independent Encodings are never concatenated.

Encoding register

Encoding PacketBig

Variant PacketBody; Contract WireShape (closed); big-endian, most-significant-first; 29 bits. Source facts: f0049, f0050, f0051, f0080, f0082.

Datagram family: WireFamily. Source facts: f0079.

Field	Shape owner	Bit range / width	Value type / presence	Source facts
PacketFlag	Header: WireShape	0–0 / 1	boolean / required	f0052, f0057, f0058, f0083
PacketNumber	Variant: PacketBody	1–12 / 12	signed / required	f0053, f0055, f0065, f0066
PacketPayload	Variant: PacketBody	13–28 / 16	bytes / required	f0054, f0056, f0067, f0068

Encoding PacketLittle

Variant PacketBody; Contract WireShape (closed); little-endian, least-significant-first; 29 bits. Source facts: f0059, f0060, f0061, f0080, f0082.

Datagram family: WireFamily. Source facts: f0079.

Field	Shape owner	Bit range / width	Value type / presence	Source facts
PacketFlag	Header: WireShape	0–0 / 1	boolean / required	f0057, f0058, f0062, f0083
PacketNumber	Variant: PacketBody	1–12 / 12	signed / required	f0055, f0063, f0065, f0066
PacketPayload	Variant: PacketBody	13–28 / 16	bytes / required	f0056, f0064, f0067, f0068

Variant coverage

A logical Variant does not define a wire layout. Missing Encoding is a modeling gap, not evidence about native codec implementation.

Variant	Contract	Encodings	Source facts
PacketBody	WireShape	PacketBig, PacketLittle	f0049, f0059, f0080
WireVariant	WireShape	No Encoding declared; no fixed wire layout	f0081